

Introduction to Electronics and Arduino!

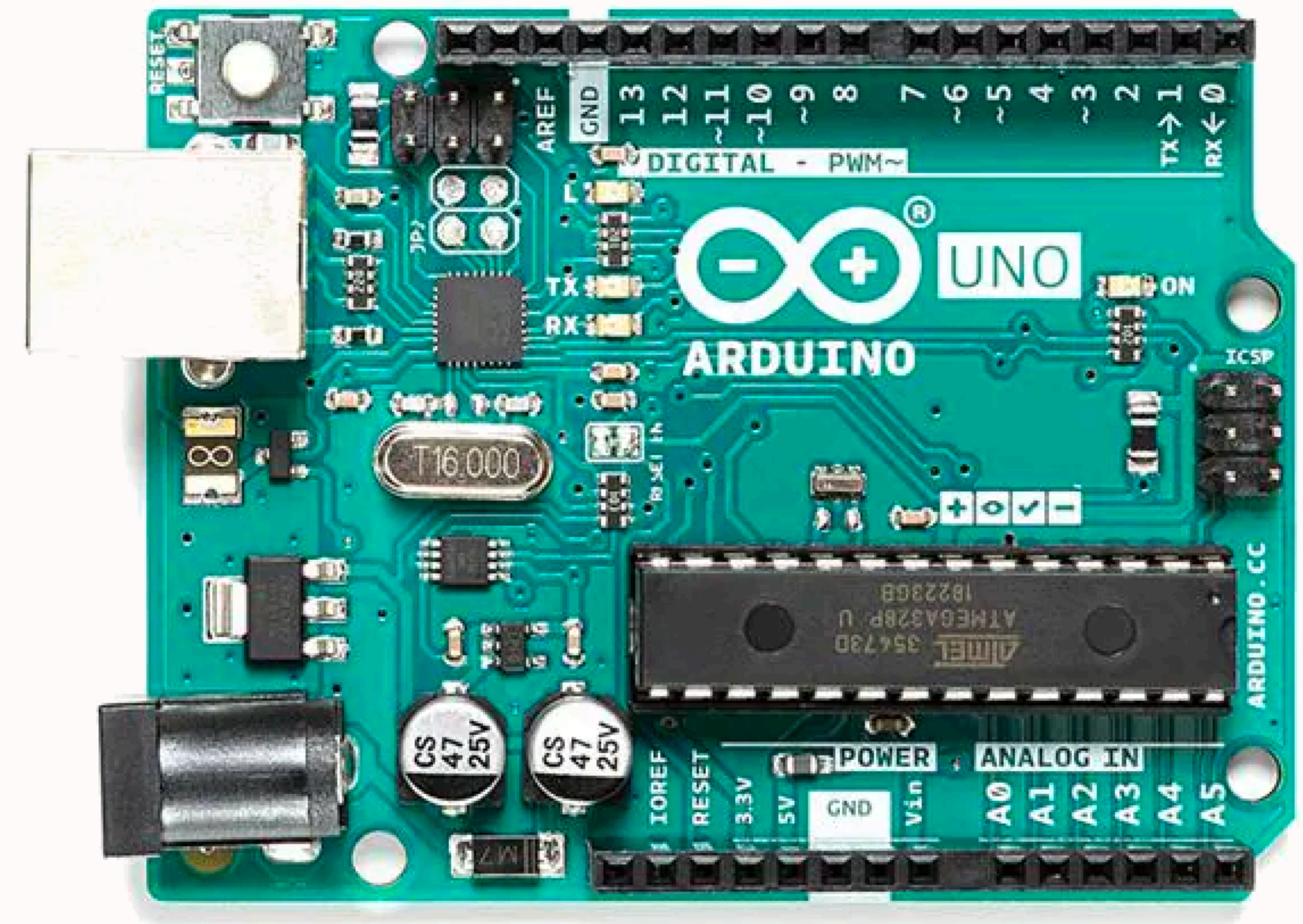
Basics of Arduino Circuits *Eric, Andy, Allen*

Introduction

- We are going over everything from the basics
- Learning *why* is just as important as *what*

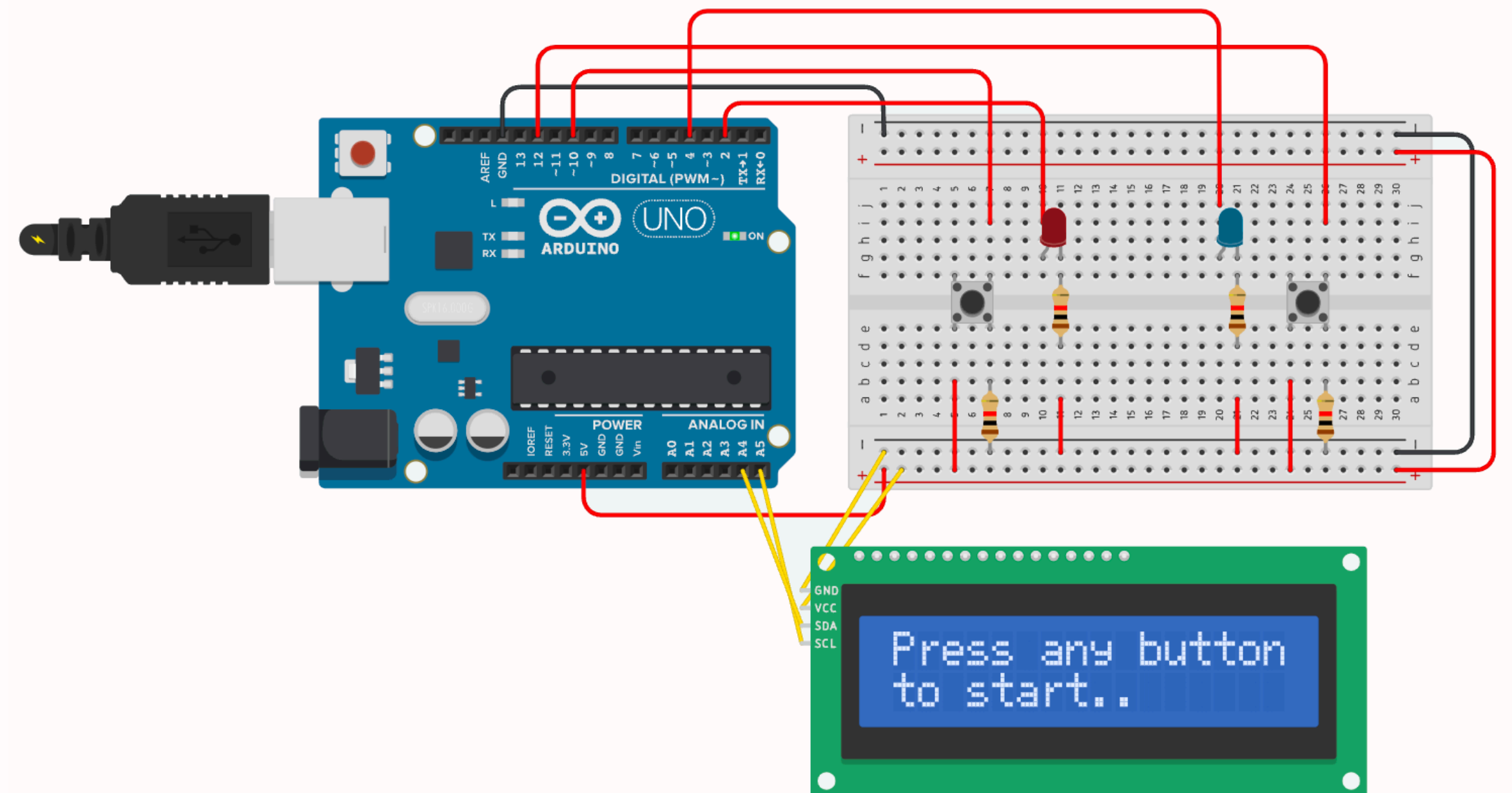
For Students

- Follow along with the workshop
- Ask questions whenever you want
- Be creative



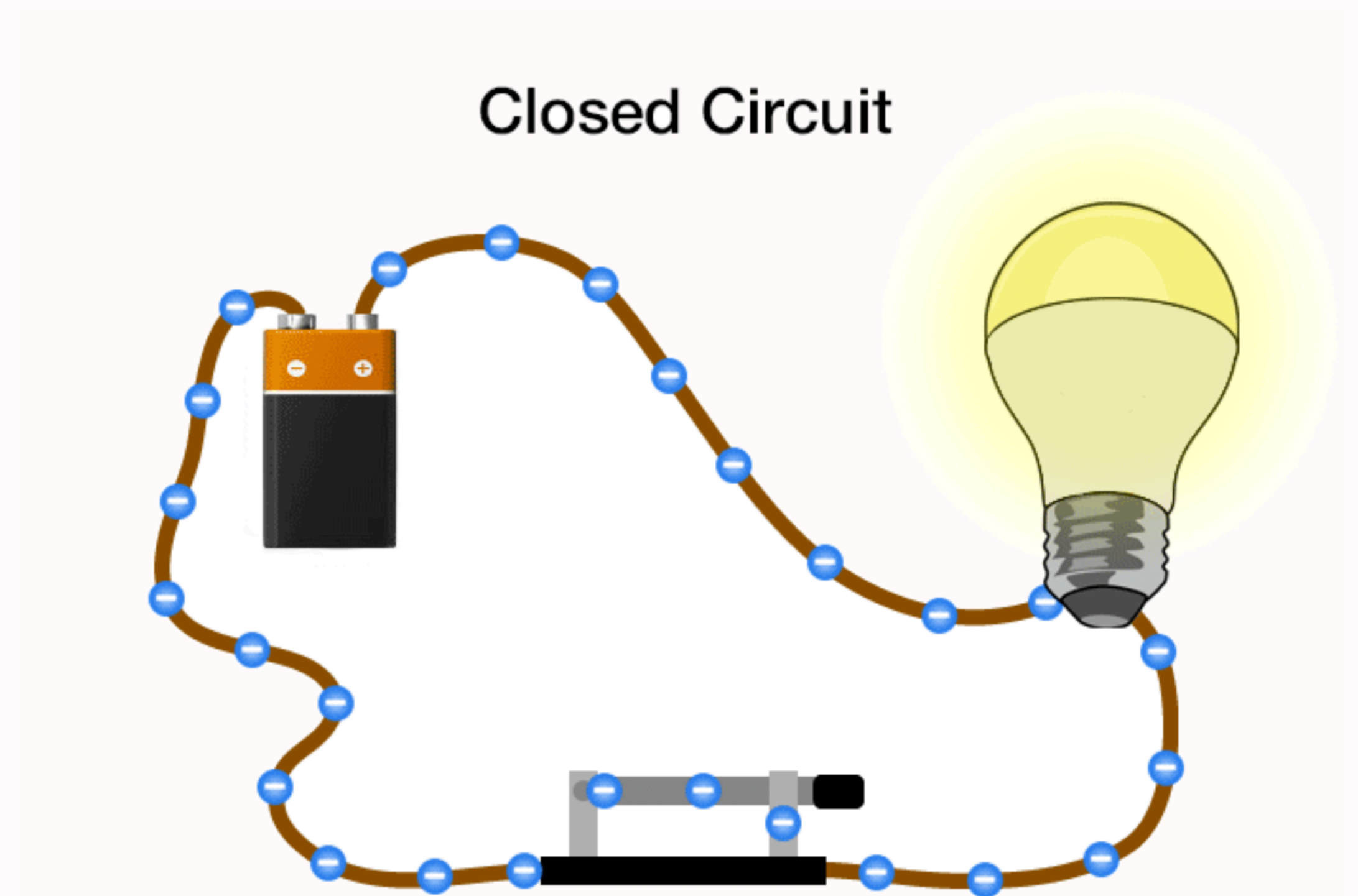
Outline of Workshop

- 1. Electricity
- 2. Circuits
- 3. Basic Arduino Programming
- 4. Arduino with Circuits
- 5. Creating the Project

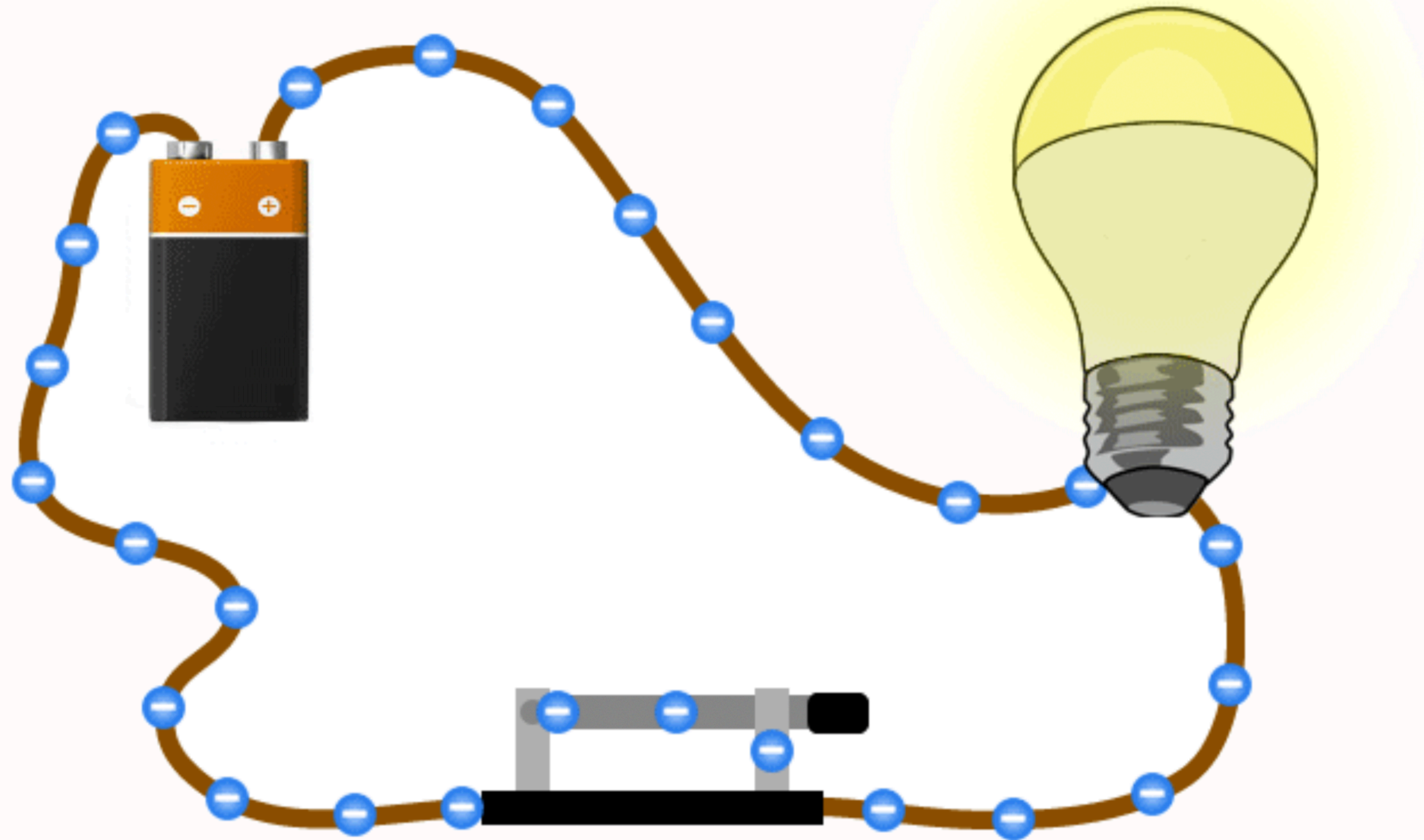


Electricity

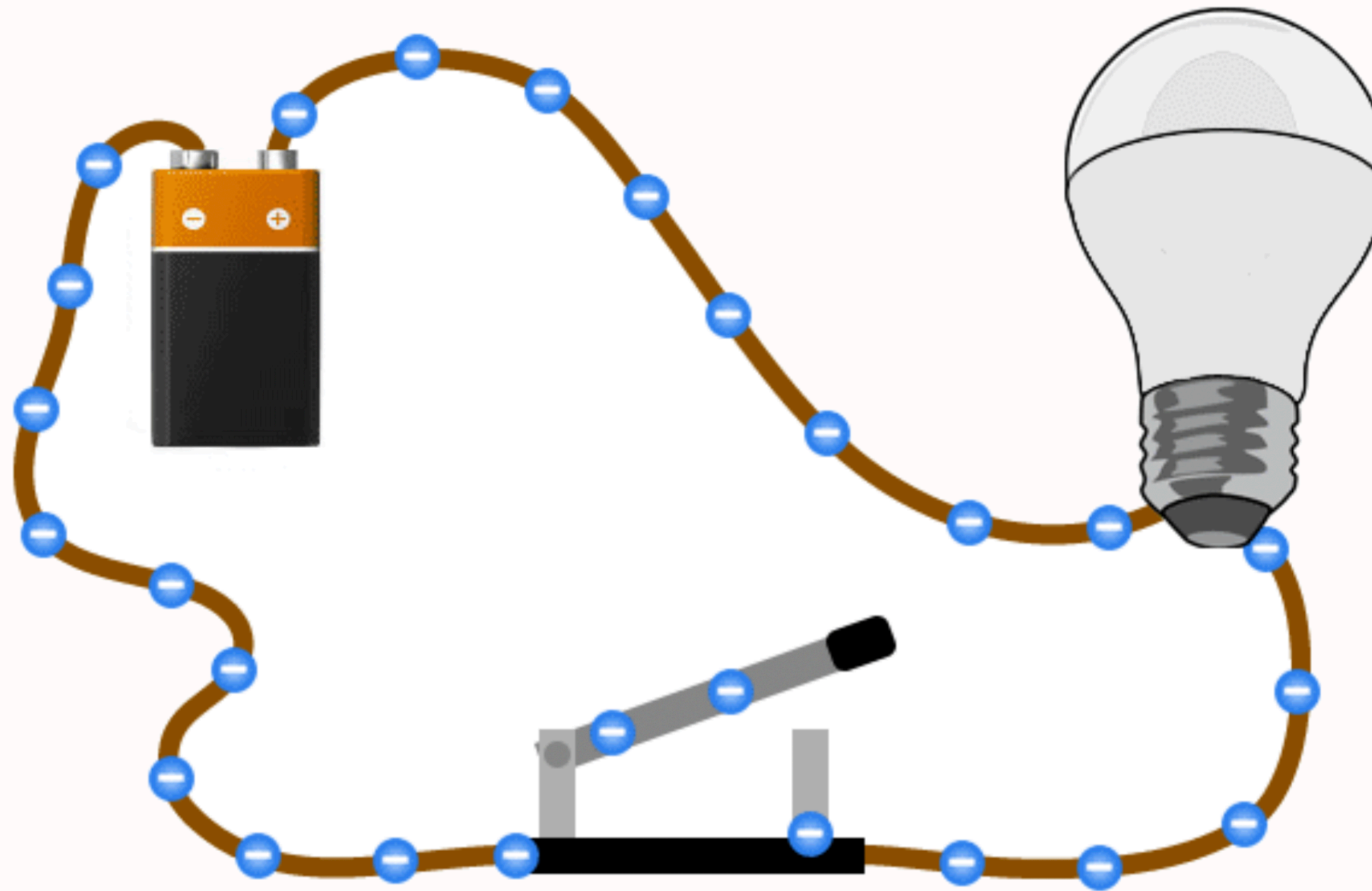
- Electrons flowing from negative to positive
- Powers components in its path (LEDs, Arduino, Light Bulb)
- Electricity can only flow if there is a path

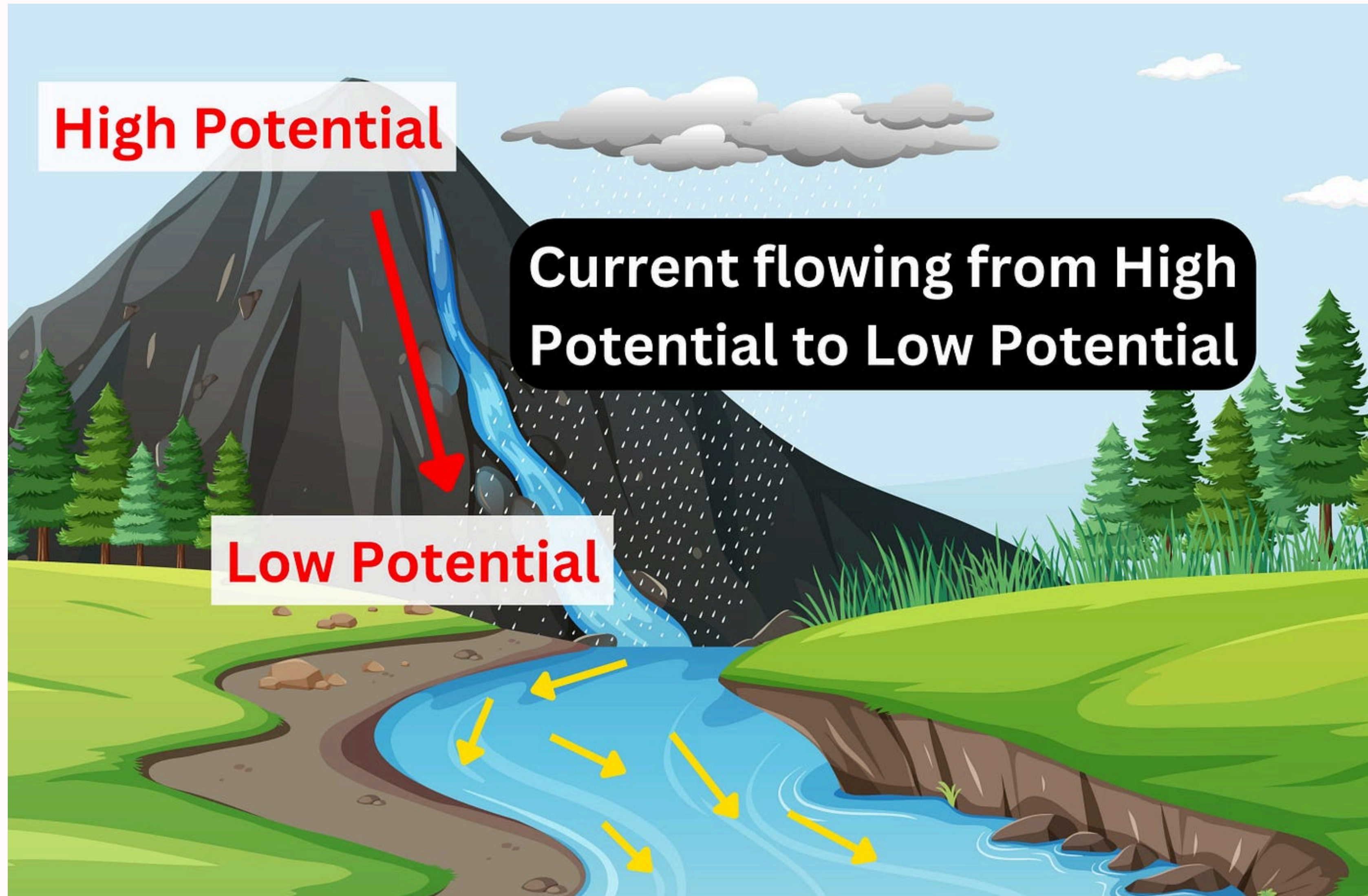


Closed Circuit



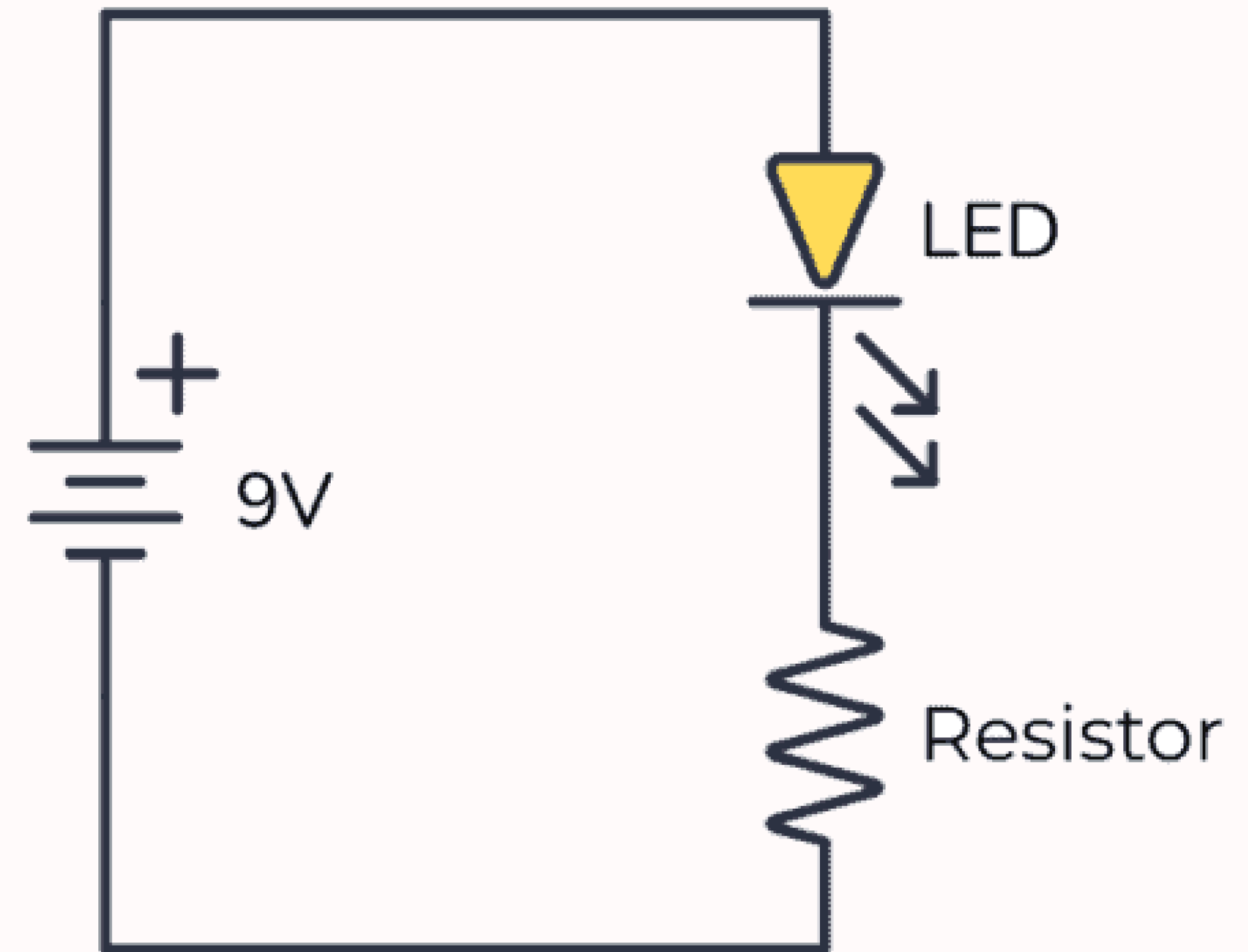
Open Circuit

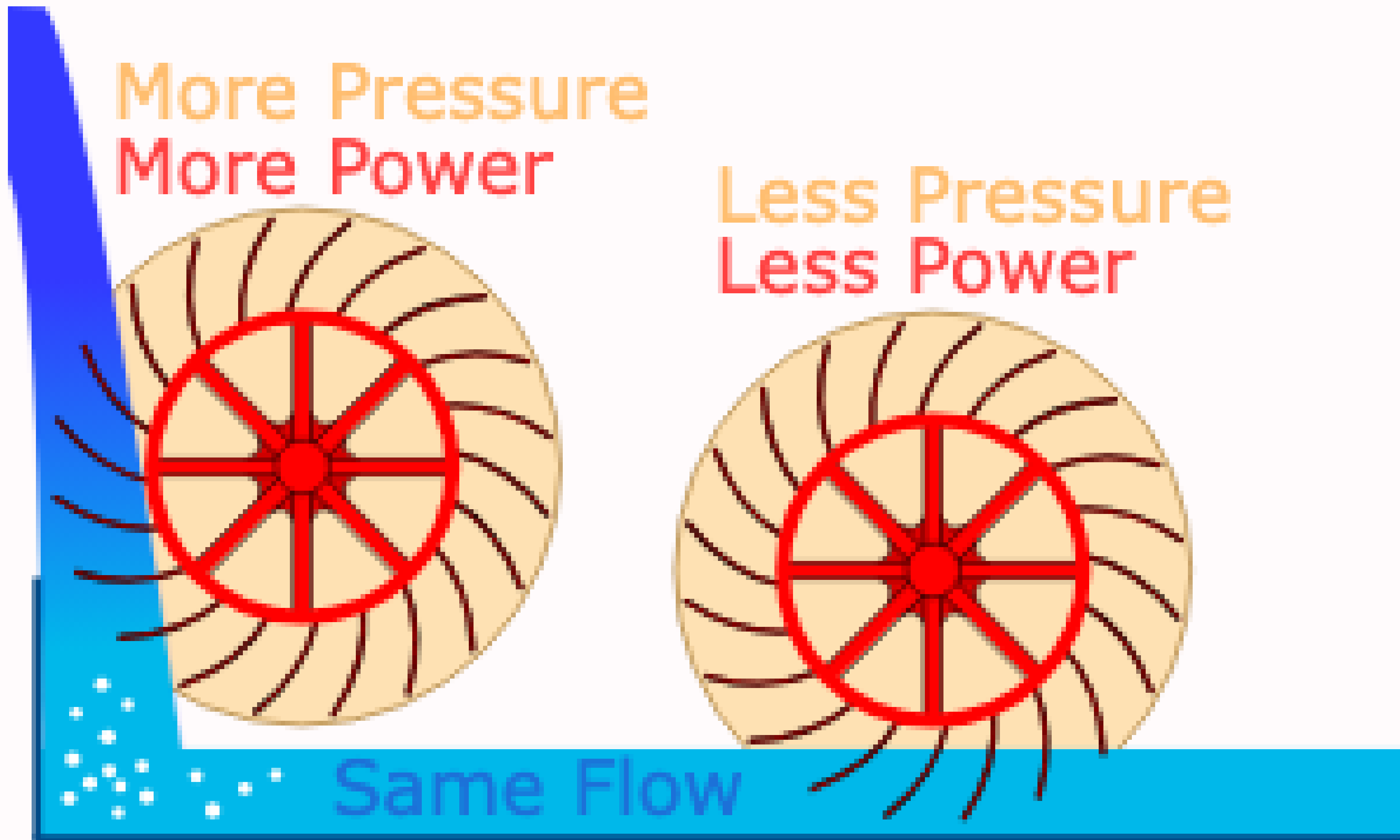




Electricity Terms

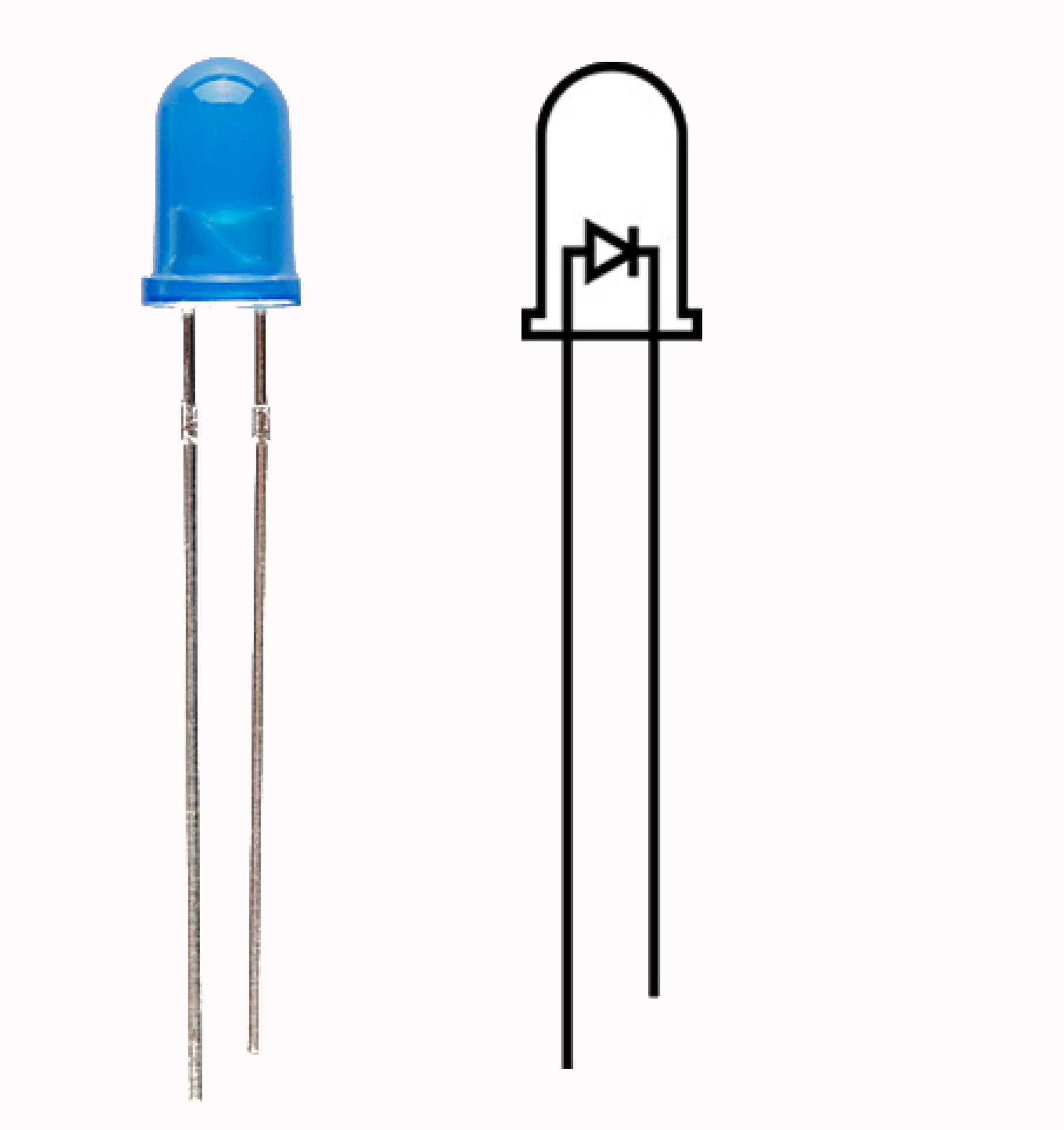
- **Current** = How much electricity flows per second
- **Voltage** = Potential electricity
- **Resistance** = How much electricity a material stops
- Ohm's Law: $V = IR \implies I = \frac{V}{R}$
- Too much current = some components overheat and break
 - LEDs, Arduino, etc.





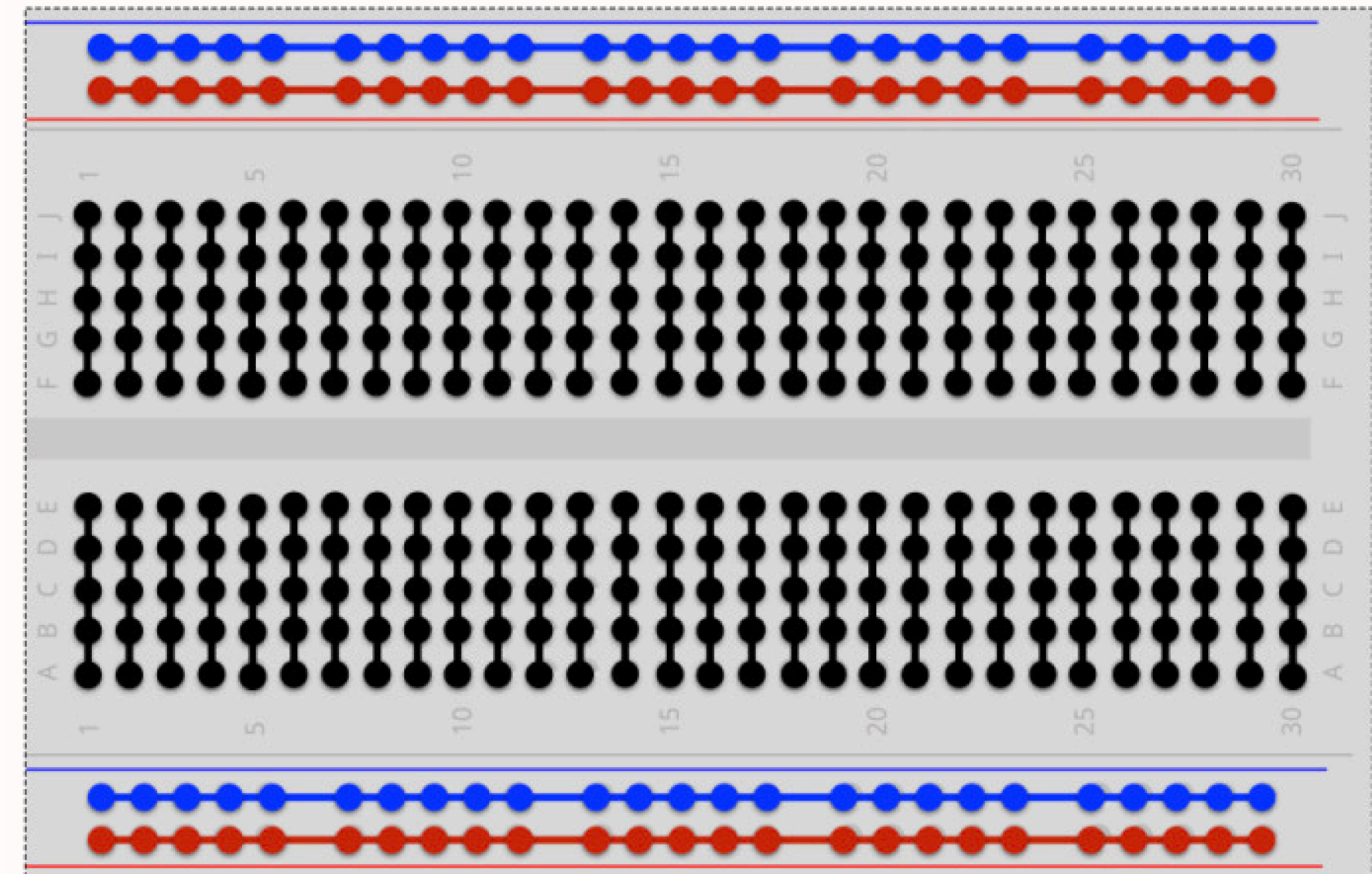
LEDs

- **Light-emitting diode**
- Diode lets current go through *one direction*
- LED = *"thing that lights up when current goes through in a certain direction"*
- Longer end to high voltage (**5V**), shorter end to low voltage (**Ground**)



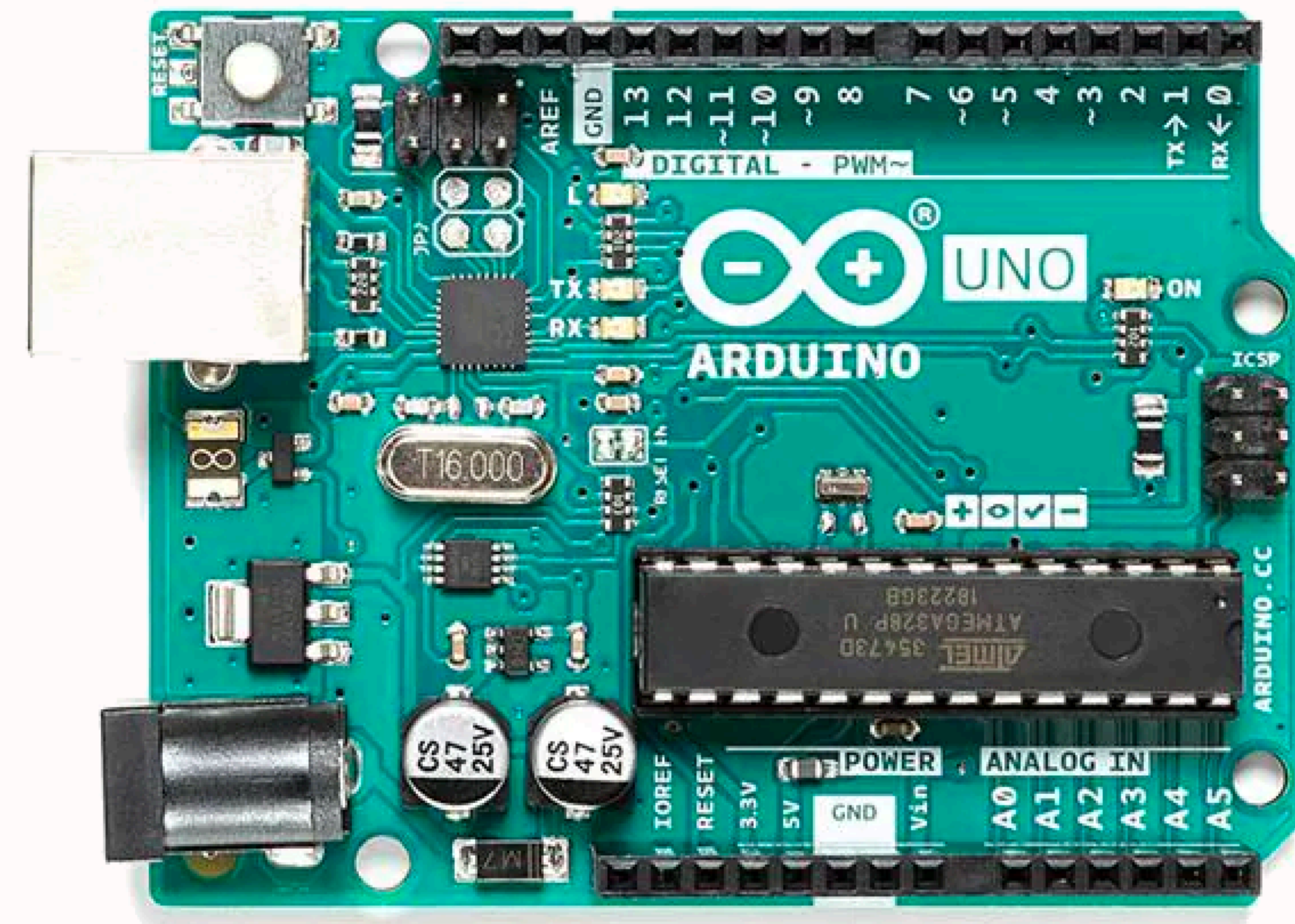
Circuits and Wiring

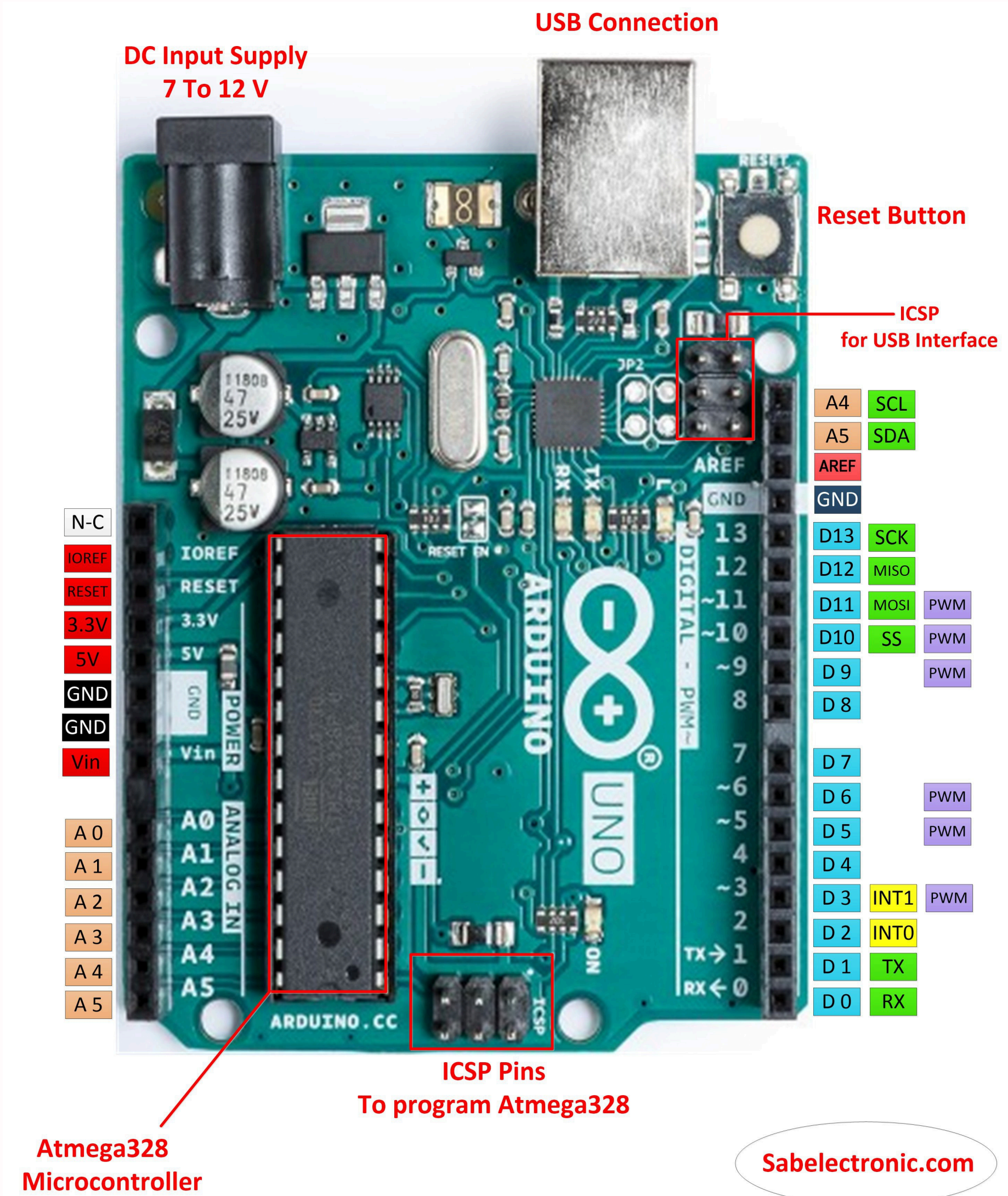
- Circuit on **breadboard**, with wires and other components
 - LEDs, Buttons, LCDs, Arduino, resistor, etc.
- Breadboards make wiring *easy to change*
 - Connects wires just by plugging in
 - Blue is connected, red is connected, black is connected
 - Great for learning & projects



Arduino

- Has pins for *output* and *input*
- Has pins for **ground** and **V5** / **V3.3** (constant)
- Connects to power source (could be computer)
- Computer connects to upload code
- Arduino runs code that is uploaded, with any power source





Arduino Code

- Uses Arduino language (C++ with special built-in functions)
 - *digitalWrite(...)*, *delay(...)*, *analogRead(...)*, etc.
- Runs **setup**, then runs **loop** function until it is off

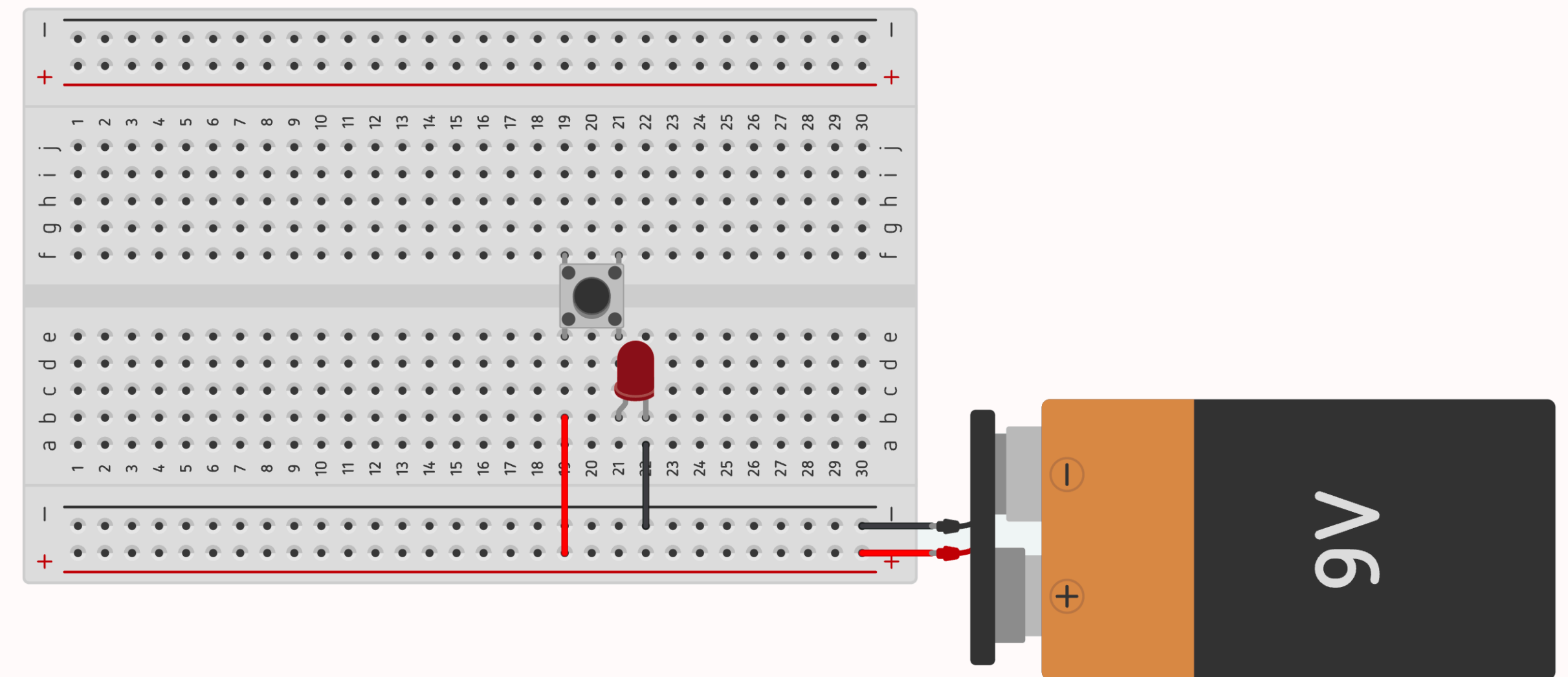
Code that turns LED on and off

```
// The setup() function runs once arduino gets power
void setup() {
    pinMode(LED_BUILTIN, OUTPUT);
}

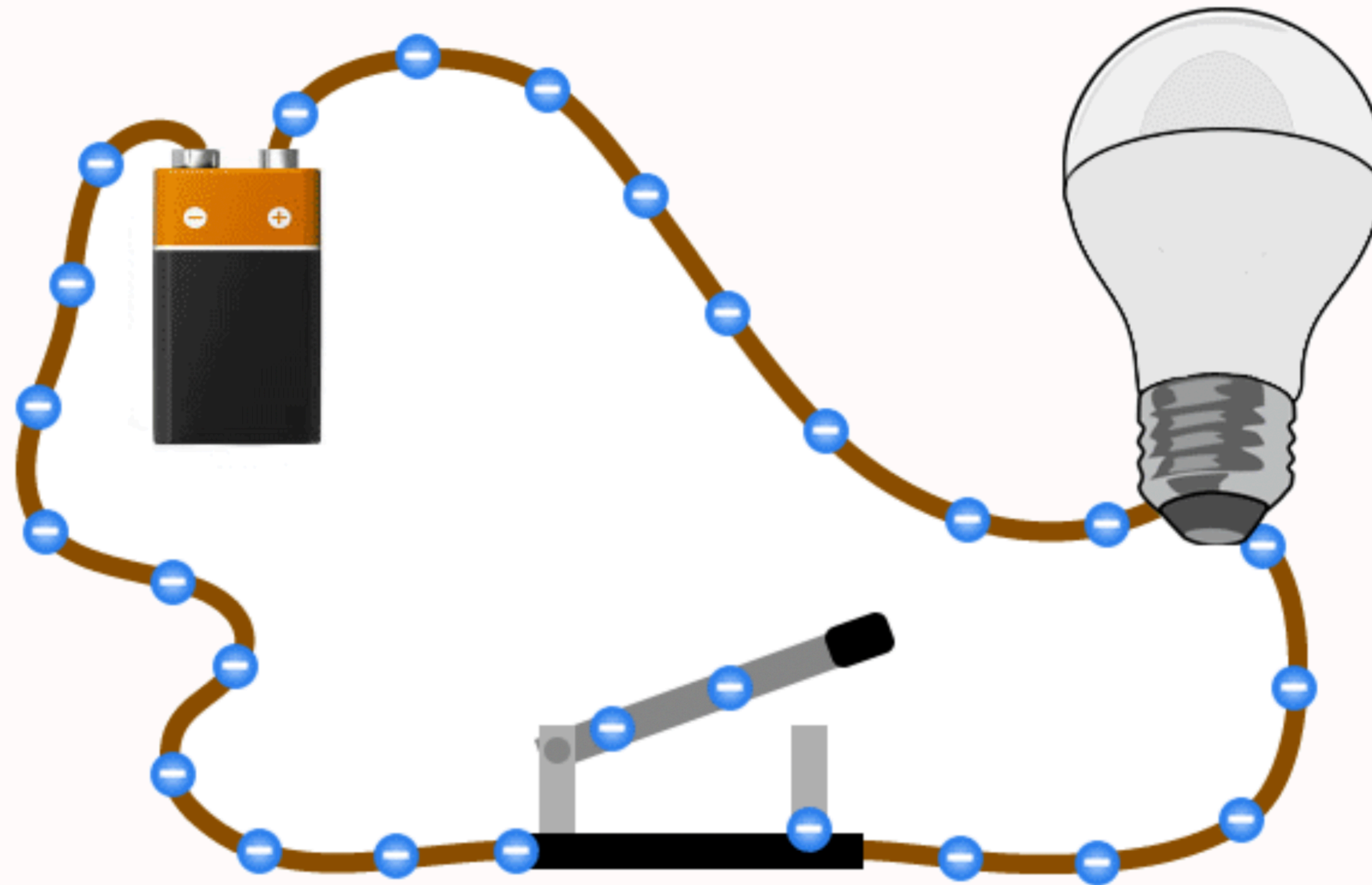
// the loop() function runs over and over again
forever
void loop() {
    digitalWrite(LED_BUILTIN, HIGH);
    delay(200);
    digitalWrite(LED_BUILTIN, LOW);
    delay(200);
}
```

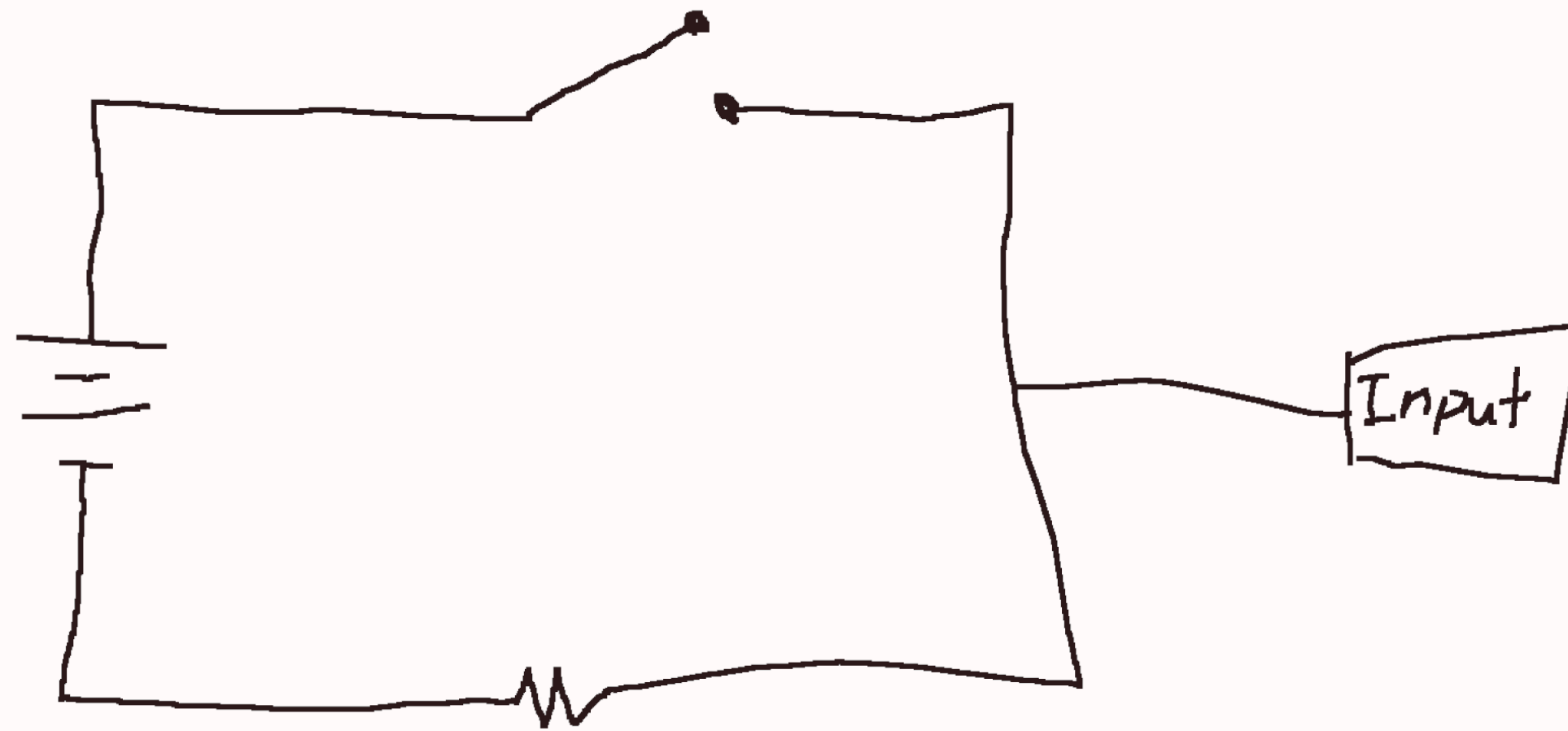
Buttons

- Buttons control when circuit is open or closed
- Pushing button connects circuit
- "Pull-Down Resistors"* are necessary when using buttons for input
 - Get rid of excess charge (sends it to **ground**)

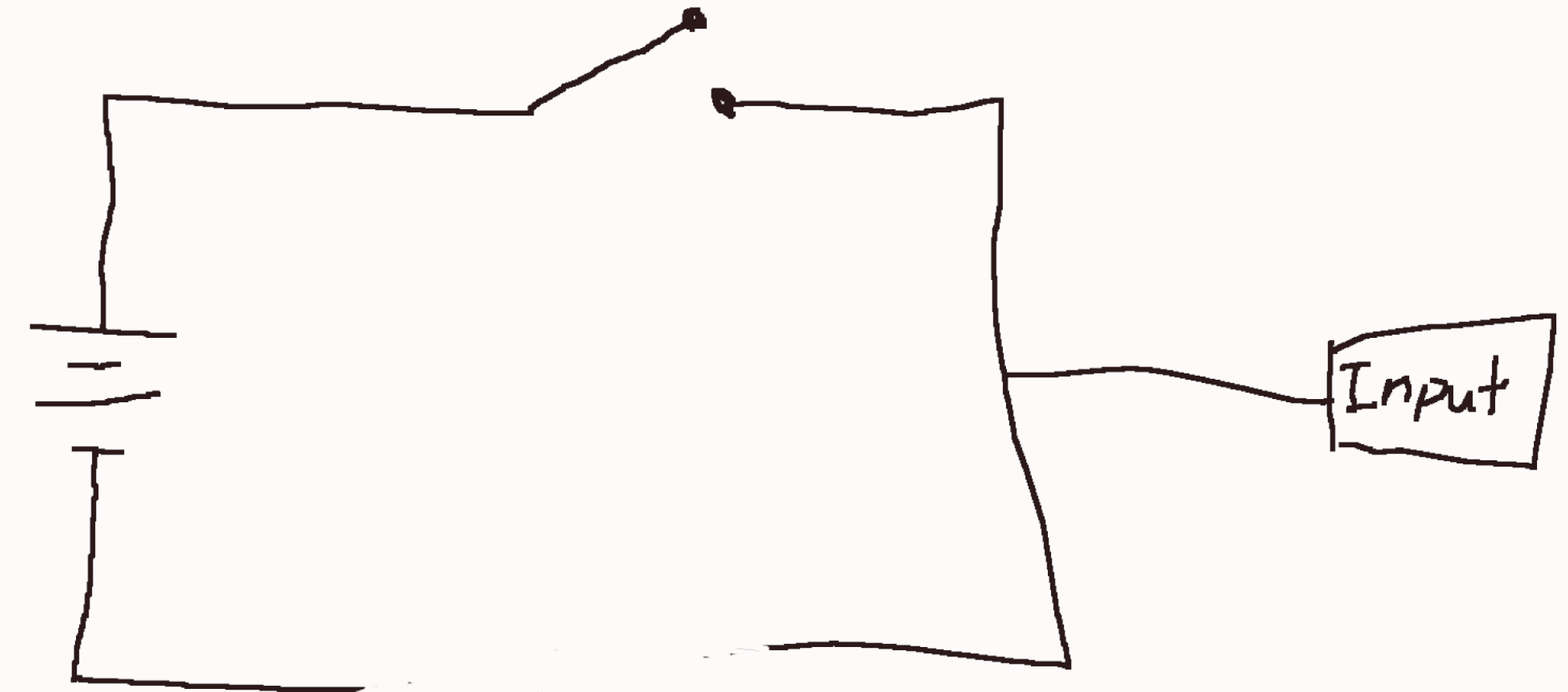


Open Circuit





(with pulldown)



(without pulldown)

(the slideshow budget ran out)

Code with Button Logic

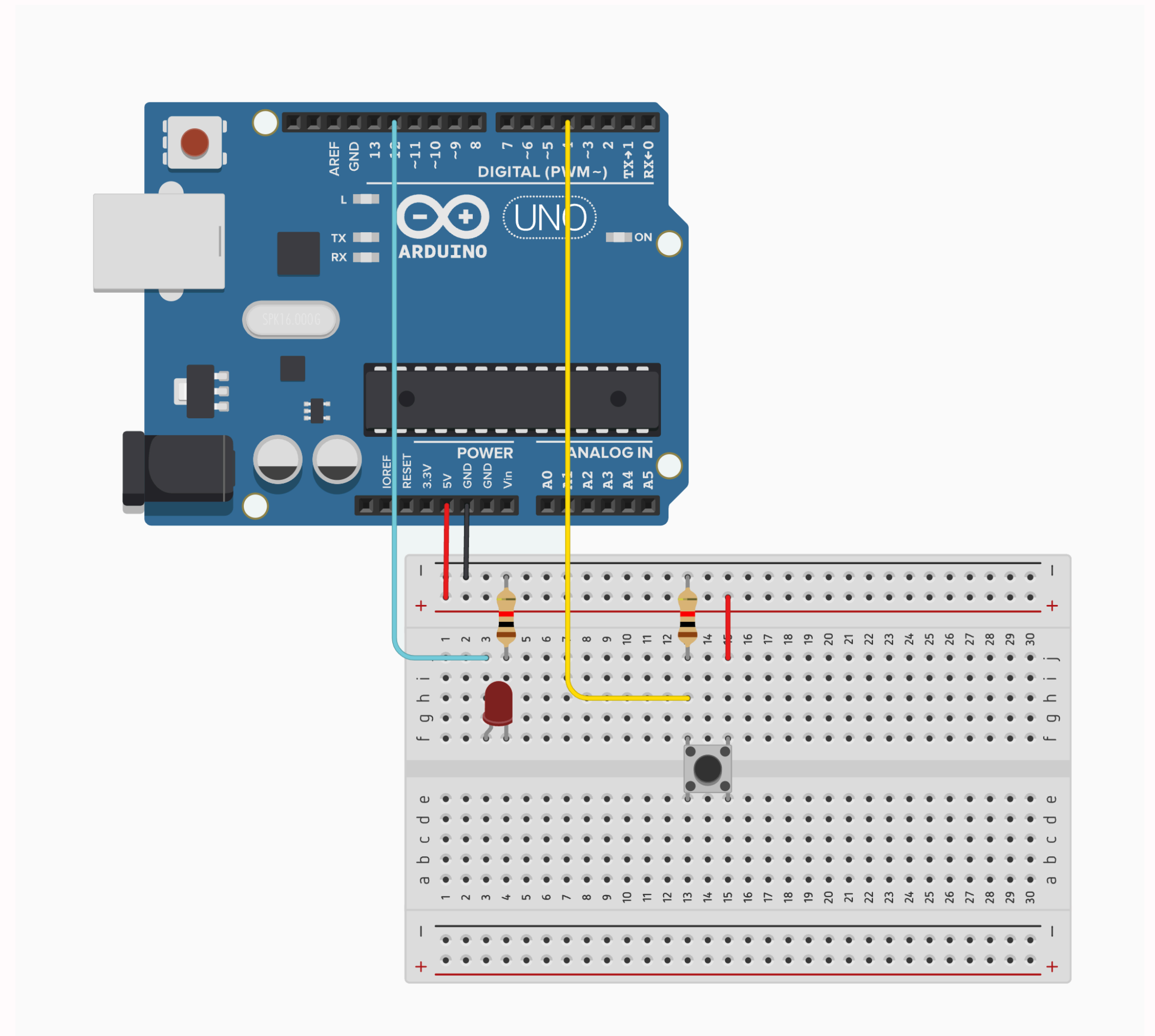
```
const int BUTTON_PIN = 10;
const int LED_PIN = 6;

void setup() {
  pinMode(LED_PIN, OUTPUT);
  pinMode(BUTTON_PIN, INPUT); // set pin to input
}

void loop() {
  // runs if there is high input to the pin
  if (digitalRead(BUTTON_PIN) == HIGH) {
    // flash light once
    digitalWrite(LED_PIN, HIGH);
    delay(100);
    digitalWrite(LED_PIN, LOW);
    delay(100);
  }
}
```

Controlling LEDs

- Connect output to LED
- Connect input to button
- Use programming to check input
- Send voltage to LED output



```
const int LED_PIN = 12;
const int BUTTON_PIN = 4;

void setup() {
  pinMode(LED_PIN, OUTPUT);
  pinMode(BUTTON_PIN, INPUT);
}

void loop() {
  if (digitalRead(BUTTON_PIN) == HIGH) {
    digitalWrite(LED_PIN, HIGH);
    delay(50);
    digitalWrite(LED_PIN, LOW);
    delay(50);
  } else {
    digitalWrite(LED_PIN, HIGH);
    delay(200);
    digitalWrite(LED_PIN, LOW);
    delay(200);
  }
}
```

Can you try with 2 LEDs and 2 Buttons?

```
const int LED_A_PIN = 12;
const int LED_B_PIN = 13;
const int BUTTON_A_PIN = 4;
const int BUTTON_B_PIN = 5;

void setup() {
  pinMode(LED_A_PIN, OUTPUT);
  pinMode(LED_B_PIN, OUTPUT);
  pinMode(BUTTON_A_PIN, INPUT);
  pinMode(BUTTON_B_PIN, INPUT);
}

void loop() {
  bool a = digitalRead(BUTTON_A_PIN) == HIGH;
  bool b = digitalRead(BUTTON_B_PIN) == HIGH;

  if (a && b) {
    digitalWrite(LED_A_PIN, HIGH);
    digitalWrite(LED_B_PIN, LOW);
    delay(50);
    digitalWrite(LED_A_PIN, LOW);
    digitalWrite(LED_B_PIN, HIGH);
    delay(50);
  }
}
```

```
    } else if (a) {  
        digitalWrite(LED_A_PIN, HIGH);  
        delay(50);  
        digitalWrite(LED_A_PIN, LOW);  
        delay(50);  
    } else if (b) {  
        digitalWrite(LED_B_PIN, HIGH);  
        delay(50);  
        digitalWrite(LED_B_PIN, LOW);  
        delay(50);  
    }  
  
    digitalWrite(LED_A_PIN, LOW);  
    digitalWrite(LED_B_PIN, LOW);  
}
```

THANK YOU FOR COMING TO THE WORKSHOP